SHM with Energy Practice Problems

1. A mass on a spring has a *period* of 0.7 seconds and an amplitude of 1.2 meters.
 (a) Write down its equation of motion $x(t)$.
 (b) Calculate the total energy of this system.
 (c) Calculate the maximum speed of the mass.

2. An object with mass 1.8 kg is attached to a spring with spring constant $k = 250$ N/m and executes simple harmonic motion. At the moment when the object is 0.030 m from its equilibrium position, its speed is 0.40 m/s. Write the equation for motion $x(t)$.